BATCHIT

Collaborative Bulk Purchasing Platform

Full System Plan & Design Document

Version 1.0

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Table of Contents

Table of Contents 2
1. Executive Summary 4
2. Problem Statement 4
  2.1 The Core Problem 4
  2.2 Problems with the Original Peer-to-Peer Model 4
  2.3 The Shifted Solution: Provider-Centric Model 5
3. System Overview 5
  3.1 Key Entities 5
  3.2 How a Batch Works 6
4. User Flows 6
  4.1 Customer Onboarding Flow 6
  4.2 Returning User Flow 6
  4.3 Creating a Batch (Customer) 7
  4.4 Joining a Batch (Customer) 7
  4.5 Provider Flow (High-Level) 7
5. Application Screens (Customer App) 7
  5.1 Screen Map 7
  5.2 Home Screen — Detailed Layout 8
  5.3 Batch Card — UI Component 8
6. Subscriptions & Provider Relationships 9
  6.1 Subscribing to a Provider 9
  6.2 Subscription Management 9
7. Notification System 9
  7.1 Notification Types 9
  7.2 Notification Page Layout 10
8. Data Models 10
  8.1 Customer 10
  8.2 Provider 11
  8.3 Product 11
  8.4 Batch 12
  8.5 Batch Participant 12
9. System Architecture 12
  9.1 Architecture Overview 12
  9.2 Technology Stack Recommendation 13
10. Security & Privacy Considerations 14

11. Phased Rollout Plan.....14
12. Open Questions & Clarifications Needed ..... **Error! Bookmark not defined.**

1. Executive Summary

Batchit is a mobile-first collaborative bulk purchasing platform that solves a common consumer pain point: purchasing minimums that exceed individual needs. By connecting consumers who want to buy portions of a bulk product, Batchit allows users to split the cost and quantity, reducing waste and saving money.

The platform is centered around Providers (retailers, wholesalers, or suppliers) who list their products. Customers find products they need, create or join Batches — group purchase requests — and coordinate purchases through the provider. This provider-centric model removes the challenge of customer-to-customer coordination across regions and retailer preferences.

> **Core Value Proposition**
>
> Batchit reduces per-unit purchase cost and product waste by enabling customers to pool demand. Providers gain access to aggregated bulk orders, improving their sales volume. Everyone wins.

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2. Problem Statement

2.1 The Core Problem

Many consumer goods are sold in bulk quantities that exceed what a single buyer needs. This creates several friction points:

- A customer needs 15 toilet rolls but the pack contains 20 — they either overspend or go without.
- Bulk packs offer lower per-unit prices, but individuals cannot always take advantage of that pricing.
- Product waste increases when buyers are forced to purchase more than required.

2.2 Problems with the Original Peer-to-Peer Model

An initial concept of connecting customers directly with each other faced the following challenges:

Problem	Impact
Hard to find others	Customers wanting to split the same product at the same time are hard to match.
Geographic barriers	Customers from different regions cannot easily coordinate a shared purchase.

Retailer mismatch	Customer A may prefer Retailer X while Customer B prefers Retailer Y — making a split impossible.
Trust issues	Peer-to-peer money handling raises concerns about fraud and reliability.
Scalability	Matching individual customers for every purchase is complex and operationally heavy.

2.3 The Shifted Solution: Provider-Centric Model

Batchit resolved these issues by pivoting to a provider-centric approach. Instead of matching customers to each other, everything revolves around a Provider — a business, retailer, or supplier who lists their goods on the platform. This means:

- All customers shopping at the same provider can join the same batch.
- Geographic friction is eliminated since the provider handles logistics.
- The provider fulfills the bulk order and manages delivery or pickup.
- Batchit facilitates the grouping and coordination, not the money transfer between users.

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3. System Overview

3.1 Key Entities

Entity	Description	Role
Customer	An individual user who wants to purchase a portion of a bulk product.	Primary user of the app
Provider	A business, retailer, or wholesaler that lists products and fulfills orders.	Supplier-side actor
Batch	A group purchase request specifying a product, total quantity, and how portions are divided among participating customers.	Core transactional object
Subscription	A relationship between a Customer and a Provider, granting access to send and receive batches from that provider.	Connectivity mechanism

3.2 How a Batch Works

A Batch is the fundamental unit of activity on Batchit. The lifecycle is as follows:

1. A customer identifies a product they want to purchase partially (e.g., 15 of 20 toilet rolls).
2. The customer creates a Batch, specifying the product, their desired quantity, and the provider.
3. The Batch is shared with the provider's subscriber pool. Other customers who want the remaining quantity can join.
4. Once the Batch is fully filled (total quantity matches the pack size), the batch is sent to the provider.
5. The provider fulfills the order in bulk. Each participant receives their purchased portion.

> **Listening & Sending — Dual Subscription Benefit**
> When subscribed to a provider, a customer can both SEND new batches to that provider AND LISTEN for batches created by other customers at that provider — so they are automatically notified and can join batches they did not create.

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4. User Flows

4.1 Customer Onboarding Flow

> **New User Journey**

6. App Launch — Splash screen with Batchit branding.
7. Onboarding Screens — 3–5 introductory slides explaining the concept: what a batch is, how providers work, and how to save money.
8. Questionnaire — Collects user preferences (product categories of interest, shopping frequency, preferred regions, budget range). Used to power recommendations.
9. Account Creation — User enters name, email/phone, and password. Optionally continue with Google/Apple.
10. Login — Authenticated session established.
11. Home Screen — User lands on the main dashboard. Profile can be completed later via the Profile section.

4.2 Returning User Flow

12. App Launch — Automatic login if session is active.
13. Home Screen — Feed of recommended batches and popular providers is shown.
14. User browses, searches, creates, or joins batches.

15. Notifications alert user to batch activity.

4.3 Creating a Batch (Customer)

16. From the Home Screen or Provider page, tap 'Create Batch'.
17. Select product from the provider's catalogue.
18. Specify desired quantity (must be less than the full pack quantity).
19. Choose which subscribed providers to send the batch to.
20. Optionally add a note or deadline for the batch.
21. Batch is published. Subscribed customers at the chosen provider are notified.

4.4 Joining a Batch (Customer)

22. User receives notification of a new batch at a subscribed provider, OR discovers a batch via search.
23. User reviews batch details: product, quantity needed, quantity available, participants.
24. User taps 'Join Batch' and specifies how many units they want.
25. If the batch is now fully filled, it moves to fulfillment.
26. If not yet full, the batch stays open for more joiners.

4.5 Provider Flow (High-Level)

While the provider side requires its own detailed specification, the high-level flow is:

27. Provider registers and creates a business profile.
28. Provider lists products: name, description, pack size, unit price, available stock.
29. Customers subscribe to the provider.
30. Provider receives incoming batches (group purchase requests).
31. Provider reviews, confirms, and fulfills the batch order.
32. Provider can also initiate special offers or bulk deals that appear as promoted batches.

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5. Application Screens (Customer App)

5.1 Screen Map

Screen	Description

Splash / Onboarding	First-time launch: brand intro, concept walkthrough, questionnaire, account creation, login.
Home Screen	Main dashboard with search bar, recommended batches, popular providers, active batches.
Search Results	Unified search returning both batches and providers filtered by query.
Provider Profile	Provider details, product catalogue, subscriber count, active batches for that provider.
Batch Detail	Full batch info, participants, quantity breakdown, join action, status.
Create Batch	Form to create a new batch: product, quantity, provider selection, notes.
My Batches	List of all batches the customer has created or joined, with status indicators.
Notifications	Chronological list of system events: new batches, batch filled, order fulfilled, etc.
Profile	Customer details, preferences, subscription management, account settings.
Provider Discovery	Browsable/searchable list of all providers with category filters.

5.2 Home Screen — Detailed Layout

The Home Screen is the primary interface. It is organized as follows, from top to bottom:

- Header — App logo (top left), notifications icon (top right), user avatar.
- Search Bar — Prominent, with toggle to switch between searching Batches vs. Providers.
- Active Batches Banner — Horizontal scroll of batches the user is currently participating in.
- Popular Providers — Horizontally scrollable cards of top-rated / most-subscribed providers.
- Recommended Batches — Vertically scrollable feed of batches matching user preferences from the questionnaire.
- Bottom Navigation Bar — Home | Search | Create Batch | Notifications | Profile.

5.3 Batch Card — UI Component

Each batch is displayed as a card with the following information:

- Product name and image.
- Provider name and logo.
- Progress bar: units filled / total pack size (e.g. 15/20).
- Number of participants.

- Price per unit (calculated from pack price divided by pack size).
- Time remaining or creation date.
- 'Join' or 'View' call-to-action button.

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6. Subscriptions & Provider Relationships

6.1 Subscribing to a Provider

Customers subscribe to providers they frequently shop at or are interested in. The subscription grants two capabilities:

Capability	What It Means
LISTEN	When any other customer creates a batch at this provider, the subscribed customer is automatically notified and can join that batch.
SEND	When the subscribed customer creates a new batch, they can choose to broadcast it to this provider so other subscribers are notified.

6.2 Subscription Management

- Customers can subscribe and unsubscribe from providers at any time via the Provider Profile or the Profile > Subscriptions section.
- Customers can control notification preferences per provider (e.g., notify for all new batches vs. only for specific product categories).
- Popular and recommended providers are surfaced on the Home Screen to encourage new subscriptions.

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7. Notification System

7.1 Notification Types

Event	Description	Channel
New Batch	A new batch has been created at a provider you subscribe to.	Push + In-App

Batch Joined	Another customer has joined your batch.	In-App
Batch Filled	A batch you are part of is now fully subscribed.	Push + In-App
Order Confirmed	The provider has confirmed the fulfilled batch order.	Push + In-App
Order Ready	Your portion of the batch order is ready for pickup/delivery.	Push + In-App
Batch Expired	A batch you were part of expired without being filled.	In-App
New Provider Deal	A subscribed provider has posted a new special offer batch.	Push (optional)
Profile Reminder	Reminder to complete profile if fields are still empty.	In-App

7.2 Notification Page Layout

- Grouped by date: Today, Yesterday, This Week, Earlier.
- Each notification shows: icon, summary text, timestamp, and a link to the relevant batch or provider.
- Unread notifications are highlighted. User can mark all as read.
- Settings shortcut from this page to manage notification preferences.

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8. Data Models

8.1 Customer

Field	Description
customer_id	UUID — Primary key
name	Full name
email	Email address (unique)
phone	Phone number (optional)
profile_photo_url	Link to avatar image
preferences	JSON — Product categories, budget range, region (from questionnaire)

subscriptions	Array of provider_ids
created_at	Timestamp
last_login	Timestamp

8.2 Provider

Field	Description
provider_id	UUID — Primary key
business_name	Display name
logo_url	Provider logo image
description	Short bio / description
category	e.g., Grocery, Electronics, Household
location	Address or service region
contact_email	Business contact
rating	Average rating (float)
subscriber_count	Cached count for display
verified	Boolean — is the provider verified by Batchit
created_at	Timestamp

8.3 Product

Field	Description
product_id	UUID — Primary key
provider_id	FK — owning provider
name	Product name
description	Product details
image_url	Product photo
pack_size	Total units per bulk pack
pack_price	Full pack price
unit_price	Calculated: pack_price / pack_size
category	Product category
in_stock	Boolean

8.4 Batch

Field	Description
batch_id	UUID — Primary key
product_id	FK — product being batched
provider_id	FK — associated provider
creator_id	FK — customer who created the batch
total_quantity	Full pack size (target quantity)
filled_quantity	How many units have been claimed
status	open | filled | confirmed | fulfilled | expired
expires_at	Optional deadline for the batch
created_at	Timestamp
notes	Optional message from creator

8.5 Batch Participant

Field	Description
participant_id	UUID — Primary key
batch_id	FK — batch
customer_id	FK — customer
quantity_requested	How many units this customer wants
joined_at	Timestamp
status	pending | confirmed | fulfilled

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9. System Architecture

9.1 Architecture Overview

Batchit follows a client-server architecture with the following major components:

Component	Responsibility

Mobile App (Customer)	React Native or Flutter app for iOS and Android. Consumes the REST/GraphQL API.
Mobile App (Provider)	Separate provider-facing app or web dashboard. Manages product listings and incoming batches.
API Gateway	Entry point for all client requests. Handles authentication, rate limiting, and routing.
Auth Service	JWT-based authentication. Supports email/password and OAuth (Google, Apple).
User Service	Manages customer and provider profiles, questionnaire data, preferences.
Batch Service	Core business logic — creating, joining, filling, expiring batches.
Notification Service	Sends push notifications (FCM/APNs) and manages in-app notification feeds.
Subscription Service	Manages customer-provider subscription relationships.
Search Service	Full-text search across batches and providers. Powered by Elasticsearch or Algolia.
Recommendation Engine	Generates personalized provider and batch recommendations based on questionnaire data and activity.
Database	PostgreSQL for relational data. Redis for caching and real-time batch state.
File Storage	AWS S3 or equivalent for product images, provider logos, and profile photos.
Push Notification Provider	Firebase Cloud Messaging (Android) + APNs (iOS).

9.2 Technology Stack

Layer	Technology
Mobile (Customer & Provider)	Flutter (shared codebase for iOS + Android)
Backend	Django Rest Framework
Database	PostgreSQL (primary), Redis (caching, session, real-time)
Search	Algolia or Elasticsearch
Authentication	JWT + Google auth
File Storage	Locally for now
Push Notifications	Firebase Cloud Messaging (FCM)
Hosting	Digital Ocean or Contabo
CI/CD	GitHub Actions or GitLab CI

Monitoring	Prometheus and Grafana, but for now no monitoring

10. Security & Privacy Considerations

- All API communication over HTTPS/TLS 1.3.
- Passwords hashed with bcrypt (minimum 12 rounds), but for now nothing just simple hashing in django
- JWT tokens with short expiry (15 minutes) and refresh token rotation.
- Customer personal data encrypted at rest (AES-256).
- GDPR / data privacy compliance — users can request data deletion.
- Provider verification process to prevent fraudulent listings.
- Rate limiting on all public-facing endpoints to prevent abuse.
- Financial transactions handled by a certified payment processor (Stripe, Flutterwave, or equivalent) — Batchit does not store card data, but for now nothing.

11. Phased Rollout Plan

Phase	Scope	Timeline
Phase 1 — MVP	Core customer app: onboarding, questionnaire, home screen, batch creation and joining, provider profiles, basic notifications.	2-3 weeks. We end here for now
Phase 2 — Provider Portal	Provider web dashboard: product listings, batch management, order fulfillment, analytics.	Months 4–7
Phase 3 — Intelligence	Recommendation engine, smart notifications, subscription listening, batch auto-matching.	Months 7–10
Phase 4 — Payments	Integrated payments: batch fee collection, refund flows, provider payouts.	Months 10–13
Phase 5 — Growth	Ratings & reviews, referral program, promotional batches, provider promotions.	Months 13–16